



Smriti (स्मृति / स्मृति) – Technical Specification

Multilingual AI-Powered Cognitive Care, Memory Companion & Clinical Monitoring Ecosystem

REACT 19 & TAILWIND V4

GOOGLE GEMINI AI

OFFLINE-FIRST PWA

MONGODB ATLAS

META WHATSAPP CLOUD API

DPDP ACT 2023

VERCEL & RENDER

App Mission & Clinical Context

Smriti is an intelligent healthcare platform built for elderly patients with Alzheimer's disease, dementia, and Mild Cognitive Impairment (MCI). Rooted in Indian and North-Eastern cultural traditions, it bridges the daily care gap between elderly patients, family caregivers, and clinicians through multi-modal voice guidance, offline PWA caching, automated WhatsApp reminders, culturally authentic cognitive games, and clinical telemetry.

1. Complete Tech Stack Master Table

Layer / Domain	Technology / Library	Purpose, Architecture & Key Implementation Details
Frontend UI	React 19.2, Vite 8, React Router v7	Ultra-responsive client SPA with sub-second HMR, route lazy loading, accessible state trees, and component isolation.
Styling & Design	Tailwind CSS v4, Lucide Icons, Canvas Confetti	Warm soothing cream palette (<code>#FAF7F2</code>), minimum 56px touch targets, high contrast, tactile feedback, and celebration animations.
Backend Engine	Node.js, Express.js (v5.x), Node-Cron	RESTful API layer, JWT middleware, WebAuthn challenge verifier, automated reminder cron engine, and webhook receivers.
Database Layer	MongoDB Atlas, Mongoose ODM (v9)	Cloud document database hosting Patients, Caregivers, Daily Reminders, Reminiscence Photos, and Cognitive Game Session logs.
Generative AI	Google Gemini AI SDK (<code>@google/genai</code>)	Powers the in-app assistant with real-time memory grounding (patient medications, schedules, family details, and empathetic dementia care dialogue).
Voice & Speech	Web Speech Synthesis & Recognition API	Hands-free microphone voice input for queries and automated natural audio read-aloud (<code>[🔊 Listen]</code>) in Hindi, Assamese, and English.
WhatsApp Messaging	Meta WhatsApp Business Cloud API (<code>wa.me</code>)	Direct WhatsApp bot link (<code>https://wa.me/15556680031</code>) and webhook router for automated medication reminder check-ins.
Offline PWA & Storage	<code>vite-plugin-pwa</code> , Workbox, IndexedDB (<code>idb</code>)	Full offline shell caching, standalone installation (iOS/Android/Desktop), local storage for profiles/photos, and resilient background sync queue.
Auth & Sessions	JSON Web Tokens (JWT), <code>bcryptjs</code>	365-day persistent JWT sessions stored in <code>localStorage</code> with isolated multi-role credentials (Patient & Caregiver).
OAuth & Biometrics	<code>@react-oauth/google</code> , WebAuthn (FIDO2)	Google OAuth 2.0 single sign-on and 1-tap hardware biometric fingerprint/Face ID authentication without password typing.
Localization	<code>i18next</code> , <code>react-i18next</code>	Full internationalization covering English, Hindi (<code>hi</code>), and Assamese (<code>as</code>) across all dashboards and games.
Telemetry & Analytics	Recharts, REST Telemetry Endpoints	Clinician performance trend charts, reaction speeds, and cognitive level progression logs.
Privacy & Governance	DPDP Act 2023 Compliance	Consent checkboxes for photos, AES-256 encrypted storage, 30-day deletion guarantees, and Grievance Officer (<code>gopalrajlohiya2@gmail.com</code>).
Cloud Hosting	Vercel (Frontend) & Render (Backend)	Single-page rewrites on Vercel and zero-downtime Node.js web services with MongoDB Atlas connection on Render.

2. Core System Architecture & Features

AI ASSISTANT

 **Google Gemini AI & Voice Assistant**

- Official Gemini Sparkle:** High-visibility AI gradient floating launcher with glowing aura.
- Voice-First STT & TTS:** Built-in microphone with real-time speech-to-text and automatic spoken audio read-aloud.
- Clinical Memory Grounding:** Backend feeds patient vitals, active prescriptions, and daily routines as live context to Gemini.

PWA & SYNC

 **Offline-First PWA & IndexedDB**

- Workbox Service Worker:** Standalone app installable on mobile and desktop devices.
- IndexedDB Local Storage (idb):** Caches patient records, photos, and 10 daily routines offline.
- Background Action Queue:** Offline routine ticks and game scores automatically synchronize with MongoDB when reconnected.

MESSAGING & CRON

 **WhatsApp Meta Cloud API & Cron**

- 1-Tap WhatsApp Bridge:** Dedicated green launcher opens direct conversation with Smriti Bot (`wa.me/15556680031`).
- Scheduled Cron Engine:** Node-cron runs daily to evaluate routine timelines and send WhatsApp reminders.
- Webhook Router:** Inbound message receiver processes patient responses and logs adherence to MongoDB.

SECURITY & AUTH

 **WebAuthn & 365-Day JWT Auth**

- FIDO2 Biometrics:** Fingerprint / Face ID login using hardware public-key cryptography (biometrics stay on-device).
- Persistent 365-Day Sessions:** Resilient session persistence prevents elderly users from unexpected logouts.
- Accessible Senior PIN Pad:** 4-digit oversized keypad with tactile tap feedback and demo shortcuts.

3. Culturally Attuned Cognitive Games Suite (5 Games)

Game Title	Cognitive Domain	Cultural & Regional Context	Clinical Telemetry & Mechanics
Market Day Basket (बज़ारबा आछि / बाज़ार की टोकरी)	Categorization, Visual Recall & Mental Math	10 North-East produce items (King Chilli, Assam Tea, Bamboo Shoot, Kaji Nemu, Lai Xaak, etc.).	5-level adaptive difficulty scaling item counts, running live score, and accuracy metrics.
Daily Routine Sequencer (दैनिक दिनचर्या क्रम)	Executive Function & Logical Sequencing	Daily living activities (brewing morning tea, watering Tulsi plant, morning bath).	Calculates step ordering logic, hesitation time, and error correction rates.
Faces & Family Recall (चेहरे और परिवार की यादें)	Reminiscence Therapy & Facial Memory	Family portraits, relationships (Grandson, Daughter, Spouse), and ancestral memories.	Tracks relationship recognition accuracy and cognitive recall speed.
Sound & Rhythm Match (ध्वनि और लय मिलान)	Auditory Perception & Pattern Recall	Traditional instruments (Bihu Dhol, Pepa flute, Temple Bell, Shankha, Mridangam).	Auditory sequence repetition and rhythm timing matching.
Odd One Out (अलग पहचानें)	Semantic Anomaly Discrimination	Indigenous flora/fauna, cultural festivals, and traditional kitchen items.	Category boundary identification and visual discrimination latency.

4. Clinical Caregiver Command Center

- **Multi-Patient Profile Support:** Robust resolution for distinct patient profiles (e.g. Ramesh Sharma in Guwahati, Meera Baruah in Shillong, Biren Das in Jorhat).
- **Real-Time Red Flag Alerts (/caregiver/notifications):** Instant notifications for missed medications with a 1-tap patient calling button and persistent database dismissal.
- **7-Day Adherence Charts & Routine Manager:** Visual graphs displaying medication and hydration compliance with full 10-slot routine customizers.
- **Cognitive ML Telemetry Hub (/caregiver/patient/:id/games):** Deep longitudinal score tracking and cognitive trajectory graphs for doctors and clinicians.

5. Data Privacy, Legal Governance & DPDP Act 2023

DPDP 2023 Compliance Architecture

- **Mandatory DPDP Photo Upload Consent:** Explicit checkbox requiring caregiver confirmation under the Digital Personal Data Protection Act 2023 before any senior photo is saved.
- **Comprehensive Privacy Policy Route (/privacy-policy):** Public legal page detailing data types (health notes, audio logs, photos), purpose limitation, and storage safeguards.
- **Data Security & Retention:** AES-256 database encryption at rest, HTTPS in transit, role-based caregiver access, and a 30-day data deletion guarantee upon patient request.
- **Appointed Grievance Officer:** Direct compliance contact available at `gopalrajlohiya2@gmail.com`.

6. Cloud Deployment Architecture (Vercel & Render)

Deployment Target	Configuration Files	Build, Runtime & Environment Variables
Frontend (Vercel)	<code>vercel.json</code> , <code>frontend/vercel.json</code>	• Build Command: <code>npm run build</code> (Vite 8) • Output Directory: <code>frontend/dist</code> (or <code>dist</code>) • Rewrites: Single-page rewrite <code>/(.*) → /index.html</code> • Env Variables: <code>VITE_API_URL</code>, <code>VITE_GOOGLE_CLIENT_ID</code>
Backend (Render)	<code>render.yaml</code> , <code>backend/server.js</code>	• Runtime: Node.js (v18+) web service • Start Command: <code>npm start</code> (<code>node server.js</code>) • Env Variables: <code>MONGO_URI</code>, <code>JWT_SECRET</code>, <code>GEMINI_API_KEY</code>, <code>GOOGLE_CLIENT_ID</code>, <code>WHATSAPP_TOKEN</code>

7. Key REST API Endpoints Reference

Module	Endpoint	Functionality
Caregivers	<code>POST /api/caregivers/signup</code> <code>POST /api/caregivers/login</code> <code>POST /api/caregivers/google-login</code>	Clinician registration, password auth, and Google OAuth 2.0 token verification.
Patients	<code>GET /api/patients</code> <code>POST /api/patients/login</code> <code>POST /api/patients/biometric-login</code> <code>GET /api/patients/:id/photos</code>	Patient roster listing, PIN keypad login, WebAuthn FIDO2 verification, and DPDP photo gallery.
Reminders	<code>GET /api/reminders/alerts</code> <code>PATCH /api/reminders/:id</code> <code>PATCH /api/reminders/:id/dismiss</code>	Fetch real-time Red Flag alerts, toggle completion ("Mark Done"), and dismiss alerts.
Telemetry & AI	<code>POST /api/game-sessions</code> <code>POST /api/chatbot/message</code> <code>POST /api/whatsapp</code>	Record cognitive game round telemetry, Google Gemini conversation, and Meta WhatsApp webhook.